EKSPEDISI AI NUSANTARA 2026

"Grounding the Cloud: Membumikan Kecerdasan Awan"

National Survivability and Sovereignty in the Age of Agentic AI and Emerging AGI

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DOCUMENT INFORMATION

Property	Details
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Date	December 19, 2025 (Final, Fully-Cited Master Revision)
Affiliation	Tsinghua SEA (TSEA) & United in Diversity (UID)
Minimum Budget Requirement	IDR 6,600,000,000 (excluding TSEA-X platform development) [file:162]
Campaign Duration	January – December 2026 (Deployment Phase) + January 2027 (Summit) [file:162]
Geographic Scope	38 Indonesian Provinces [file:162]
Target Participants	100,000+ AI-literate citizens across industry verticals [file:162]
Strategic Thesis	"Grounding the Cloud" — moving beyond accessing AI technology to applying it at scale across Indonesia's real economy in the wake of Agentic AI and emerging AGI [file:162]

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1. EXECUTIVE SUMMARY

The year 2026 marks the "Deployment Horizon" for the global economy. As Artificial Intelligence matures from experimental cloud-based models to practical, **agentic systems** capable of autonomous reasoning and problem-solving, the challenge for nations has fundamentally shifted [web:136]. The question is no longer *"Can we access AI technology?"* but rather *"Can we apply it safely, productively, and equitably at scale?"*

For Indonesia, this moment presents a defining strategic imperative: we must take the high-level potential of the "AI Cloud" and ground it firmly in the reality of our "Nusantara Soil"—transforming abstract digital capabilities into tangible economic tools that serve our 286.7 million citizens and amplify our 68.2% working-age population advantage [file:162].

Ekspedisi AI Nusantara 2026 is the strategic vehicle designed to accomplish this mission. Initiated by Tsinghua Southeast Asia Center and United in Diversity, this is a **National Deployment Strategy** that directly operationalizes Indonesia's **National Strategy for AI (Stranas KA 2045)** [file:163]. We will "Ground the Cloud" by converting borderless AI capabilities into contextualized solutions that solve real problems for civil servants, farmers, SME operators, and startup founders across all 38 provinces [file:162].

To achieve this, we introduce a new, research-based pedagogical architecture: the **NUSA Framework (National Upskilling Sprint for AI)**, powered by the **IRIS Cycle (Immersion-Reflection-Iteration-Scale)**. This model moves beyond traditional academic training to deploy rapid, challenge-based sprints that produce **SFIA-mapped, ISO 17024-ready certified practitioners** prepared for the age of Agentic AI. The IRIS cycle is conceptually grounded in established learning science, integrating Kolb's experiential learning, Revans' action learning, and Lean Startup's build-measure-learn loop [web:30][web:27][web:55].

Our methodology creates a synchronized feedback loop spanning three temporal phases [file:162]:

1. **THE CLOUD (June 2026):** Convening global technology leaders, national policymakers (Kemenkomdigi, KORIKA), and academic institutions in Bali to ratify the **"Nusantara AI Protocol"**—a harmonized standard for curriculum, ethics, and deployment aligned with Pancasila values [file:163]. This phase launches the **"Nusantara Digital Torch"**, creating a unified signal for national transformation.

2. **THE GROUNDING (July–December 2026):** Executing a six-month relay across all 38 provinces, deploying the **NUSA Framework** via the T6 (TSEA-X) platform. We employ a **Triple-to-Hexa Helix** grounding strategy, ensuring the national protocol is transmitted directly to local governments, campuses, and SMEs before escalating to community and financial sectors for sustainable ecosystem building.

3. **THE HORIZON (January 2027):** Culminating at the National AI Summit in Jakarta, where aggregated IRIS projects generate the **"Peta Potensi AI Nusantara"** (National AI Potential Map)—a sovereign, data-driven intelligence asset visualizing our nation's human capital distribution and investment opportunities [file:162].

The campaign operates under a lean, sustainable budget model with a baseline of IDR 6.6 billion (USD $400K), delivering [file:162]:

- **100,000+ certified practitioners** mapped to the **SFIA global digital skills framework (Level 3/4)**, ready for both AI-augmented employment and SME entrepreneurship [web:145][web:127].
- 38 provincial relay events with standardized **NUSA/IRIS** delivery.
- **Blockchain-verified credentials** aligned with international **ISO/IEC 17024** standards for certification of persons [web:157][web:150].
- National media visibility through MURI record attempts [file:162].
- Crowdsourced AI potential intelligence enabling sovereign strategic planning.

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2. STRATEGIC CONTEXT & REGIONAL LANDSCAPE

2.1 The 2026 Deployment Horizon: Why Now?

Southeast Asia's AI ecosystem is entering a hyper-acceleration phase, projected to generate USD 835 billion in economic value by 2030, with compound annual growth exceeding 25% [file:162]. However, a critical bottleneck threatens this opportunity: across ASEAN, approximately 30% of AI-critical technical roles remain unfilled due to insufficient domestic talent pipelines [file:162]. In Indonesia specifically, the nation faces a need for up to 9 million digital professionals by 2030 to fully realize its digital economy potential [web:70].

The challenge is not technology access; it is **deployment capability**. Without immediate intervention, Indonesia risks a "K-shaped" AI adoption trajectory where elite institutions capture gains while rural regions and SMEs are left behind, widening the economic inequality chasm [file:162][web:140]. Ekspedisi AI Nusantara 2026 is the strategic intervention designed to invert this trajectory through mass, distributed upskilling.

2.2 Indonesia's Positioning Within ASEAN: The "Application Engine" Strategy

Indonesia operates within a competitive ASEAN AI ecosystem [file:162]. While Singapore positions itself as the **"R&D Hub"** and Vietnam as the **"Coding Factory,"** Indonesia's optimal

strategy is to become the **"Application Engine."** This strategy leverages our three distinct competitive advantages:

1. **Demographic Scale & Diversity:** With 286.7 million citizens and 42.4% under age 25, we have the raw human capital to develop millions of AI-literate practitioners [file:162].
2. **Problem Density & Economic Complexity:** Our diverse economy—with 66 million SMEs, a vast agricultural sector, and a complex maritime landscape—is an ideal laboratory for impactful AI solutions [file:162].
3. **Market Access & Regional Influence:** As ASEAN's largest economy, Indonesia is the natural test market for AI solutions at scale [file:162].

Strategic Positioning: Indonesia's advantage lies in **Scale and Application**. Ekspedisi AI Nusantara 2026 activates this by building 100,000 AI-literate practitioners, shifting Indonesia from a consumer of foreign AI products to a producer of indigenous AI solutions.

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3. ALIGNMENT WITH STRANAS KA 2045 & THE AGI HORIZON

[INSERT DIAGRAM: The National Alignment Stack (Stranas -> Ekspedisi -> Impact)]

3.1 Operationalizing the National Strategy

Ekspedisi AI Nusantara 2026 is explicitly designed to operationalize Indonesia's **National Strategy for Artificial Intelligence (Stranas KA 2045)** [file:163]. We translate the strategy's four national missions into concrete operational engines:

Stranas KA 2045 Mission	Ekspedisi AI Nusantara Implementation
Ethics & Policy	**Nusantara AI Protocol** — A harmonized standard ensuring Trustworthy AI aligned with Pancasila values and national resilience principles [file:163].
Talent Development	**NUSA Framework + IRIS Cycle** — A sprint model targeting **SFIA Level 3–4 competencies**, preparing talent for both industry roles and entrepreneurship, fulfilling the Stranas mandate [file:163][web:127].
Infrastructure & Data	**T6 Platform + Peta Potensi AI Nusantara** — A sovereign digital backbone and

Stranas KA 2045 Mission	Ekspedisi AI Nusantara Implementation
	intelligence asset enabling data sovereignty, as envisioned by Stranas [file:163].
Research & Innovation	**Hexa-Helix Challenges** — Demand-driven innovation sprints that catalyze the **KORI-KA** (Kolaborasi Riset dan Inovasi Kecerdasan Artifisial) ecosystem at the provincial level [file:163][web:74].

3.2 The "Long Game": Sovereignty, Resilience, and the AGI Horizon

Consistent with Stranas KA 2045, we proactively prepare for the **AGI Horizon**. The strategy document explicitly states, *"Since the beginning, it is necessary to consider the direction of AGI development for national interests"* [file:163]. We address this through three pillars of sovereignty:

1. **Talent Sovereignty (Human-in-the-Loop Governance):** As AI systems become agentic, human oversight is critical [web:136]. The NUSA curriculum embeds **Human-in-the-Loop (HITL)** and **Human-in-Command (HIC)** principles, as defined in Stranas [file:163]. We train Fellows not just to build AI, but to govern it.

2. **Data Sovereignty (Sovereign Intelligence):** The **Peta Potensi AI Nusantara** generates a domestically owned, real-time map of national AI capability. This reduces reliance on foreign consulting firms, ensuring the state possesses the intelligence needed to navigate the AGI transition independently.

3. **Infrastructure Resilience (Survival AI):** The T6 platform employs a **"Survival AI"** design philosophy (Offline-First). This ensures that AI capability reaches every island, regardless of connectivity, protecting equity of access and resilience against external dependencies [file:162].

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4. THE SOLUTION ARCHITECTURE: "GROUNDING THE CLOUD"

Our methodology orchestrates seamless coordination across three distinct temporal phases [file:162].

Phase 1: THE CLOUD (Strategic Alignment & Establishment) | June 2026

Venue: AISEA Bridge Conference (AISEABC), UID Bali Campus

The campaign begins at the "Cloud Layer," convening the Triple-Helix sectors to establish foundational standards:

Objective 1.1: Ratify the "Nusantara AI Protocol" A harmonized standard establishing:

- **Curriculum Standardization:** Mapped to the **SFIA** global standard [web:145].
- **Ethical Frameworks:** Localized AI governance reflecting Pancasila [file:163].
- **Certification Standards:** Aligned with **ISO/IEC 17024** for certification of persons [web:157].

Objective 1.2: Launch the "Nusantara Digital Torch" A symbolic and operational mechanism for transmitting the Protocol to provincial hosts.

[INSERT DIAGRAM: AISEA Bridge / Phase 1]

Phase 2: THE GROUNDING (Operational Deployment) | July – December 2026

Vehicle: Ekspedisi AI Nusantara—Provincial relay across all 38 provinces **Methodology: NUSA Framework** powered by the **IRIS Cycle**

The "Grounding" Transmission Mechanism (Triple-to-Hexa Helix) We transmit the high-level Protocol to the ground through specific channels:

Helix	**National Source (Cloud)**	**Provincial Target (Ground)**	**NUSA Sprint Activity**
Government	Kemenkomdigi, KORIKA	**Pemda / ASN**	AI for Public Service & Governance
Academia	Top Universities (ITB, UI)	**Local Campus / Students**	Curriculum Adoption & SFIA Certification
Business	Tech Giants, Kadin	**Local SMEs / UMKM**	Digital Transformation & AI Tooling

Helix	National Source (Cloud)	Provincial Target (Ground)	NUSA Sprint Activity
Community	NGOs, Civil Society	**Local Communities**	Inclusive Access & Digital Literacy
Media	National Press	**Regional Media**	Storytelling & "Digital Torch" Amplification
Finance	OJK, BI, VCs	**BPD / Local Banks**	Funding Promising Prototypes

The NUSA Framework: 4-Week Sprint Model

- **Day 1:** Intensive In-Person Bootcamp (Immersion).
- **Weeks 2-4:** Online T6 Platform Learning (Reflection, Iteration, Scale).
- **Capacity:** 2,500–3,000 per province (100,000+ total) [file:162].

The IRIS Cycle: The Pedagogical Engine Inspired by *irisan* (intersection), IRIS turns local problems into competencies. It integrates experiential learning [web:30], action learning [web:27], challenge-based learning [web:40], and Lean Startup principles [web:55].

[INSERT DIAGRAM: NUSA + IRIS Cycle inside T6]

1. **IMMERSION (Week 1):** Teams immerse in real challenges from local stakeholders and map them to **SFIA targets**.
2. **REFLECTION (Week 2):** Participants analyze context using action learning and self-assess against SFIA descriptors [web:125].
3. **ITERATION (Week 3):** Teams build functional prototypes using **Build-Measure-Learn** loops on T6's AI toolchain.
4. **SCALE (Week 4):** Final validation and presentation. Participants receive their **ISO 17024-ready Credential**.

Phase 3: THE HORIZON (Aggregation & Legacy) | January 2027

Venue: National AI Summit, Jakarta

Deliverable 3.1: The "Peta Potensi AI Nusantara" A dynamic dashboard aggregating IRIS project data to reveal talent distribution and sectoral innovation.

Deliverable 3.2: MURI Records "Longest Relay AI Conference" and "Most Blockchain-Verified Credentials Issued" [file:162].

Deliverable 3.3: The "Nusantara AI Fellowship" Establishment of the Fellowship as a permanent institution for continuous AI talent development [file:162].

[INSERT DIAGRAM: Three Phases + T6 Backbone]

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5. TECHNICAL INFRASTRUCTURE: THE T6 PLATFORM

[INSERT DIAGRAM: T6 AI -> Peta Potensi AI Nasional]

The T6 (TSEA-X) platform is the digital backbone, engineered for sovereignty and resilience.

5.1 Core Capabilities

- **Native AI Environments:** Cloud-hosted Jupyter notebooks, LLM interfaces, and Low-Code tools accessible via browser.
- **Socratic AI Tutor:** An AI pedagogical agent guiding learners through the IRIS cycle.
- **SFIA Mapping Engine:** Tags learner progress against SFIA descriptors [web:141].

5.2 Offline-First Architecture ("Survival AI") T6 employs a **"Survival AI"** philosophy, using PWA technology and sync queuing to ensure learning continues regardless of infrastructure limitations [file:162].

5.3 Blockchain Credential Verification Each IRIS project completion generates a blockchain-verified credential, mapped to SFIA and aligned with ISO 17024 principles, ensuring trust and portability [file:162][web:157].

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6. ACCREDITATION & STANDARDS STRATEGY

To ensure international trust and high value, we align with global standards.

1. **SFIA (Skills Framework for the Information Age):** We use SFIA, the global framework used in 200+ countries, as our competency spine [web:145][web:148]. All outcomes are mapped to **SFIA Level 3 (Apply)** and **Level 4 (Enable)**, making our graduates globally legible [web:125][web:127].

2. **ISO/IEC 17024 Alignment:** The **Nusantara AI Fellow** certification scheme is **designed to be ISO/IEC 17024-ready**. We maintain organizational separation between training and certification, ensuring impartial, competence-based assessment and paving the way for formal accreditation [web:157][web:150].

3. **Dual-Path Impact Strategy:** The NUSA curriculum prepares Fellows for two outcomes:

    - **AI-Augmented Worker:** Ready for employment in modern firms.
    - **AI-Enabled Entrepreneur:** Ready to launch new ventures or upgrade SMEs.

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7. STRATEGIC OUTCOMES & ECONOMIC PROJECTIONS

7.1 Peta Potensi AI Nusantara A sovereign intelligence asset providing real-time data on talent clusters and sectoral innovation, replacing assumption-based planning with evidence.

7.2 Economic Impact Metrics

- **Job Creation:** Injecting 100,000+ ready-to-work practitioners into the economy. Research suggests AI will drive 170 million new jobs globally by 2030 [web:131].
- **SME Productivity:** Delivering cost reductions and efficiency gains for participating SMEs.
- **New Ventures:** Creating a pipeline of AI-native micro-businesses, contributing to the 26% GDP growth AI is projected to add globally [web:134].

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8. IMPLEMENTATION ROADMAP

- **January 2026:** Mobilization & Partnerships.
- **February 2026:** Curriculum Development & Beta Testing.
- **March 2026:** Institutional Confirmation & MURI Pre-approval.
- **April–May 2026:** Logistics, Media Prep, & Cabinet Endorsement.
- **June 2026: THE CLOUD** (Bali Conference & Launch).
- **July–September 2026: THE GROUNDING** (Provincial Relay).
- **October–December 2026:** Consolidation & Data Aggregation.
- **January 2027: THE HORIZON** (Jakarta Summit & Legacy).

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9. BUDGET ALLOCATION

Total Baseline: IDR 6.6 billion (USD $400,000) *Excluding Platform Dev* [file:162]

Budget Category	Allocation	Details
T6 Platform Operations	IDR 1 billion	Cloud infrastructure, API costs, blockchain fees.
Provincial Delivery	IDR 3.8 billion	Instructor honorariums, logistics, venue rentals.
AISEA Bridge Conf.	IDR 250 million	Bali venue, catering, livestream production.
Media & MURI	IDR 500 million	Documentation, verification, press materials.
National AI Summit	IDR 750 million	Jakarta venue, visualization infra, logistics.
Contingency	IDR 300 million	Operational contingency.

Note: Certification design and SFIA mapping governance are included in operational allocations.

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10. PARTNERSHIP & GOVERNANCE STRUCTURE

The campaign operates through a **Hexa-Helix collaboration framework** [file:163][web:74]:

- **Initiators (TSEA & UID):** Strategy, Platform, Financial Management.
- **Academia (Led by ITB/UI): Curriculum & Certification Authority.** Maintains NUSA/IRIS pedagogy and ISO 17024-aligned standards.
- **Government (Led by KORIKA): Policy Patronage.** Ensures alignment with Stranas KA 2045.
- **Industry (Led by Telkom): Market Drivers.** Providing real-world challenges and hiring pipelines.
- **Civil Society: Inclusivity Watchdogs.** Ensuring equitable access.
- **Media: Amplifiers.** Storytelling and Digital Torch visibility.
- **Finance: Enablers.** Funding for scalable prototypes.

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11. RISK ASSESSMENT & MITIGATION

Risk Category	Risk Description	Mitigation Strategy
Enrollment Risk	Insufficient participants in remote provinces.	Government mandate; community partnerships.
Technology Risk	T6 failures or connectivity issues.	**Survival AI (Offline-First)** architecture [file:162].
Quality Risk	Inconsistent training quality.	Standardized **IRIS Cycle**; SFIA-based rubrics.
Certification Risk	Misalignment with international standards.	Establish **NUSA Standards Council**; design for **ISO 17024** from Day 1 [web:157].
Funding Risk	Budget overruns.	Partner cost-sharing; lean operational model.

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12. SUCCESS METRICS

Participation:

- **Total Participants:** 100,000+
- **Diversity:** ≥45% women; ≥30% underrepresented regions [file:162].

Learning Outcomes:

- **IRIS Project Quality:** ≥70% achieve functional prototype validation.
- **SFIA Competency:** ≥70% demonstrate Level 3 capability; ≥20% reach Level 4 [web:127].
- **Entrepreneurial Readiness:** ≥20% of projects show viable SME potential.

Economic Impact:

- **Employment:** ≥60% employed in AI-related roles within 12 months [file:162].
- **New Ventures:** ≥10 AI startups and ≥200 AI-enabled SME initiatives.

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13. STAKEHOLDER VALUE PROPOSITION

- **For Government:** Access to an **SFIA-mapped, ISO 17024-ready** workforce and sovereign intelligence data (**Peta Potensi**), directly executing Stranas KA 2045 [file:163].
- **For Industry:** Access to talent with globally legible (SFIA) profiles, reducing recruitment costs and de-risking investment.
- **For Academia:** Curriculum innovation and credit-transfer pathways via SFIA/e-CF equivalence mapping [web:158].
- **For Society:** Equitable access to the AI economy, fostering both employment and entrepreneurship across all 38 provinces.

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14. CONCLUSION

Ekspedisi AI Nusantara 2026 is a nation-building initiative. By operationalizing the **NUSA Framework** and **IRIS Cycle**, aligned with **SFIA** and **Stranas KA 2045**, we move beyond the hype of the Cloud and plant the seeds of AI capability firmly in the Ground of Nusantara.

We build a workforce that is skilled, certified, and ready—not just to consume AI, but to master it. We prepare Indonesia not just for the digital economy of today, but for the **Agentic Future** of tomorrow.

The path forward is clear. Let us ground the cloud.

APPENDICES

Appendix A: Key Terminology Guide

- **NUSA Framework:** National Upskilling Sprint for AI.
- **IRIS Cycle:** Immersion-Reflection-Iteration-Scale.
- **Nusantara AI Fellow:** A certified practitioner (SFIA L3/4).
- **Peta Potensi AI Nusantara:** The sovereign national map of AI talent.
- **Hexa-Helix:** Collaboration model (Govt, Academia, Industry, Community, Media, Finance).
- **SFIA:** Skills Framework for the Information Age.
- **ISO/IEC 17024:** The international standard for bodies operating certification of persons.
- **Stranas KA 2045:** Indonesia's National Strategy for Artificial Intelligence.

Appendix B: Stranas-Aligned Challenge Catalogue (Summary)

Examples of IRIS challenges aligned with National Strategy sectors [file:163]:

- **Agro-Maritime:** Precision fertilizer recommendation (SFIA Data Analysis L3).
- **Public Services:** Multilingual government chatbot (SFIA Software Dev L3).
- **Healthcare:** Telemedicine triage assistant (SFIA Data Analysis L3).
- **SME:** Inventory optimization for Warung (SFIA Data Analysis L2).